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H8SCOOP

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BITS & PIECES

More for your money. again. For those who did not notice, about 3/4 through last year I changed the format of H8SCOOP giving more characters per column which resulted in a gain of about 1/2 to 1 extra printed page in the 8 page issue. With the DIABLO 630 printer's ELITE type, again, you are getting more print for your money. With this latest change, you are getting about 3 extra pages in this issue than in early last years 8 page issues. I will continue to find ways to better H8SCOOP.

MORE FEEDBACK ON CP/M

Walt Bilofsky writes in his opinion on CP/M vs. HDOS. Last month I printed two views, this is still another.

Walt says "I have more respect for the clean engineering and friendliness of HDOS. But CP/M has some features that are irresistible. Batch mode is one. Another, which is extremely important on 5" systems, is the way Heath implements "phantom drives", allowing you to group a file or run a program off a "nonexistent" drive with minimum hassle and no mount/dismount commands. I would still recommend HDOS for a beginner, but, especially for a real "hacker", there's a lot to like about CP/M. Finally, I'd like to observe that whichever system you're used to, you're not going to dislike the other one for a while once you start using it. You've got to give it a chance."

NIX ON CHEAP DUAL H77

Ray Albrektson writes in "I tried ordering all the housing parts for a H-77 dual drive from Heath. I got a semi-polite notice back that they were only replacing damaged parts and was my H-77 damaged? That nixed that idea for building a housing for some extra drives at a cheaper rate than Heath's H-77." All I can say is WOW.

ON TED BENGLER'S VOICE SYNTHESIZER

Ted told me to pass along the word that those who purchased the first cassette tape demo of the talking H8 can exchange it without cost for a new version which demonstrates the voice synthesizer he is now selling. The main difference is inflection in the voice, which makes it sound more human. If you want to "hear" this new synthesizer, write Ted Bengler, 822 County Rd. 30, Ft. Collins, CO 80525.

Ted sent me a unit for evaluation, and all I can say, is it was much beyond my dreams. This thing can do what a unit costing \$100,000 came close to doing 10 years ago. I'll have a report on this next month.

NO MORE AUTOSCRIBE FOR H8SCOOP

I tried AUTOSCRIBE this month for H8SCOOP, and it will be the last month. It has features I really liked, which other word processor packages do not have, but it also has drawbacks. Since I intend to do an article on word processor and text editors in the future, I won't get too long winded on this. Briefly, AUTOSCRIBE combined with a DIABLO has an output close to typesetting. Most word processors for right justification only space between words, AUTOSCRIBE not only does this, but also between letters in words, to make a much smoother look.

What I can't understand, while the DIABLO is capable of page widths to 158 characters for ELITE, and more for 15 pitch, autoscribe margins only support to 99, which makes cut and paste necessary for a newsletter such as this--a real bummer. Most word processors print one direction only, while the DIABLO has bi-direction capabilities which AUTOSCRIBE makes use of. That's a plus. I think the worse point is the difficulty of making revisions under AUTOSCRIBE.

Anyway, I checked TEXTIT, and the justification was not working right on that one, so enough of that. I checked VIDEO SCRIBE which seems pretty good, but does not have underline capabilities, and is a large .ABS file. So I ended up converting the AUTOSCRIBE files back to PIE8 and RUNOFF again for this month. See PROGRAMMING GOODIES for this trick.

ON H9 GRI KEYBOARD STUFF

From GRI (1-800-445-5218) per February '81 H8SCOOP comes the following prices. Wally Izuo, a H8SCOOP subscriber passed these along to me.

AR-1 Auto Repeat for GRI keyboard no.756, \$10. AL-1 Alpha Lock for GRI keyboard no. 756, \$10. 756MF mounting frame for GRI keyboard no. 756, \$5.95.

ON COMPUTER CONTINUOUS POWER ON

I received lot of feedback on last months REQUEST asking if the computer should be kept running or shut off between use.

Bill Levin says "The failure mechanism is usually thermal stress by applying power to the equipment. Once it is up to temperature, there is relatively little stress applied to the IC's."

Larry Wier says "My system has been on for three years. I work for a large data processing group which does all the hardware maintenance. We never turn the equipment off unless it dies. All our failures have been traced to excessive temperature, brown-out, or power fail at turn on. I would suggest all equipment be left on provided a positive method of temperature control is used. My H8 is fully loaded and sets on rack slides with the complete bottom exposed to air flow. I also added a switch to the fan in the (3) disk enclosure. The fan is turned on only when high levels of disk activity are contemplated. The H19 and H9 are always on with the screen cleared and the cursor in the HOME position."

From Henry Beechhold, "When my machine's not in use, it's off. I think the initial current surge can be minimized by any one of number of line filter/ transient suppressor units on the market. I use a Blitz Bug and an ISO-1. Maybe I'm kidding myself, but so far so good."

ON COMPUTER RFI

Also feedback was received on RFI to nearby television sets. It seem some users have no interference at all, and others have much.

From Larry Wier "I have been operating the H8 within 6 feet of my HF and VHF equipment without so much as a whisper from the H8. This was not true until I made sure all H8 covers made good contact with each other. Be sure the ground pin of the NEMA power plug is connected to the same ground as the radio or TV set(s). The most noise generated on my H8 is around 29.546 MHz with lesser harmonics at X2 and X3 the 29.546 signal."

Walter Scott wrote to Heath about RFI with the H19 and they wrote back "We do have available from our Parts Department a small modification which should help eliminate some of the RFI being generated by the H19 video terminal. The amount of interference reduced by this mod varies with each installation."

Also from Walter "Most of the RFI in my case occurs on TV ch 2-5 and seems to come from the H8 and H19. The TV antenna filter described in the January '81 issue of BYTE (p 68) eliminates almost all of the interference on two nearby black and white sets. My old color set, located at the other end of the house isn't affected at all."

***** Robert McGrath writes in to tell me the Heath related newsletter PIP is out of business.*****

***** BEWARE of The Information Center offering a SOURCEBOOK as a complete reference to Zenith and Heath computer systems, software, and support products. I wrote in to them for info, and received nothing.

NEXT MONTH SPECIAL ANNIVERSARY ISSUE

NEW STUFF

From John Bell Engineering, POB 338 Redwood City, CA 94064, some computer related accessories for the experimenter. John has a plus and minus 12 volt power supply kit for \$22.95, a 5 volt supply for \$16.95, a solid state switch capable of switching up to 6 amps at 120 volts from a 5 volt signal--\$10.95 for one channel and \$39.95 for 4 channels. Also available is an A/D, D/A converter for use on a parallel port for \$59.95. For more info on this, write the above address.

As promised last month, some new software from Walt Bilofsky, of The Software Toolworks, 14478 Glorietta Dr., Sherman Oaks, CA 91423 (213) 986-4885.

A new video game, graphics for MYCHESS, and software for Heath and most other CP/M systems are now available from the Software Toolworks.

INVADERS, a true video game for the H89 or H8+H1 is now available. This one has to be seen to be believed. Walt sent me a photo of the screen display, and it is really neat! Alien creatures try to land; you hold them off with your space cannon as you dodge their bombs. Play it as it comes, or customize with 23 user-variable parameters to increase the challenge; even design your own graphics. 32K required, \$19.95.

Also a full H89 graphics chessboard display for the MYCHESS Championship Chess program, plus a new self-play mode. For 48K H89 or Z80 CPU; still only \$34.95. Registered owners may update for the usual \$10; don't forget \$2 postage and handling on each order.

Also, Org 0 CP/M versions of most software offered by Walt. Available on the 5" (Heath Format) CP/M or HDOS disk are PIE 1.5 (H8 CP/M version requires H8-4) for \$29.95, ED-A-SKETCH Graphics Editor \$29.95, AIRPORT Air Traffic Controller Game \$19.95, plus the above mentioned software.

Available on standard 8" CP/M-format for Heath or any other CP/M computer, or 5" (Heath Format) CP/M or HDOS disk are the C/80 compiler at \$39.95, TEXT Formatter at \$34.95, UVMAC Z80 and UVMAC 8080 Macro Assemblers at \$29.95 each, PACK/CRYPT File compression and encryption at \$24.95, and LISP interpreter which requires 48K for \$39.95. The RATFOR Structured Fortran Preprocessor (requires Microsoft Fortran) will be available in March, \$39.95.

Available only on 5" HDOS disk are the REACH Modem/File transfer program for the H89 only at \$19.95, and the Catalog System/ Utilities disk for \$24.95.

Walt says as usual updates to registered owners will be available for \$10, but this does not include HDOS owners to update to CP/M.

To order any of Walt's software, specify the program and disk format. Include \$2.00 per order for First Class postage and handling; \$3.00 if ordering 8" disks. Overseas orders add \$1.50/item Air Mail postage. California residents add 6% tax.

Another INVADERS game in MBASIC requiring H19 and 56K memory is entirely in graphics and animation, with sound for Gregg Saville's Sound board. The program will run fine without the board, there will just be no sound.

The player controls a warrior who walks, lies down, and shoots his lazer and vaporizer guns. There are spaceships, space craft, land mines, and a lunar lander to destroy. These invaders in turn destroy the warrior by firing missiles, various death rays, crashing into the warrior, and deploying land mines that explode if stepped upon. The price on this is \$17.50. If anyone is interested in a no sound version which should run in 48K, it's the same price. Contact Jerry Phelps, 6013 Innes Trace Rd., Louisville, KY 40222 to order or for more info.

XTRA SOFT is announcing a line of professional business software, designed by professionals, and for professionals. Presently, an INVENTORY MANAGEMENT SYSTEM is available which allows you to be able to keep a detailed account of over 1100 individual items. It sells for \$199.95. Most of the software can be ordered for H8-H19, or H89. Most requires Microsoft Basic. Some will run on the H9.

Another program is POINT OF SALE, which is said to eliminate the cash register, \$199.95. To come are a mailing list and an accounting package.

Under educational Software, US PRESIDENTS for \$19.95, which can be ordered for the H9. For the chemistry minded, PERIODIC TABLE for \$19.95, again, can be ordered for H9 (48K RAM). For \$25.95 is VOCABULARY BUILDER. Available soon are TYPING TUTOR, COLLEGE ENTRANCE EXAMS, GREEK and LATIN ROOTS, NUMBER BASES, and ASSEMBLER LANGUAGE teaching program.

The business software requires you to sign a license before they will ship.

For more information on these, write XTRA SOFT, POB 91063, Louisville, KY 40291.

From TIDEWATER SOFTWARE, INC., POB 4465, Virginia Beach, VA 23454 comes DISKMOD, a new utility package designed for the H8 and H89/Z89 computers with disk drive and a minimum of 16K RAM. It will operate with any version of HDOS up to and including 1.6. DISKMOD contains the following three programs:

SAVEFILE, which provides a vehicle for recovery of a disk file in the event that a hard error is encountered.

BADSECT, permits a bad sector to be logged out of the disk's free chain so that errors are not picked up during a subsequent file creation.

DISKUSE, lists the sectors used by each file on a disk to a selected device. Output is by file name and the sectors are listed in the order in which they appear in the file.

The price of this disk is \$25, add \$7.50 for the 3 ASM files.

TIDEWATER also has sophisticated DEVICE DRIVERS for the H14, LA36 (Input and Output) as an alternate terminal, LA34 (input and output), BAUDOT TTY printer (output only) and LA36 for output only.

Each device driver greatly increases flexibility to the user's system, which has a variety of SET options not found in the Heath versions. The Input/Output ones will halt print on a CTL-S from the device's keyboard. The Baudot and LA36 driver only, will halt print on CTL-S from the console. The price on these is \$20 each, which includes ASM file. For more info on these, contact TIDEWATER.

From NEWLINE SOFTWARE, POB 402, Littleton, MA 01460 comes some software for HDOS 1.6 and the H19 or H89.

VIDEO ARTIST is a program for the creation of graphics pictures on the screen. The pictures can be saved on disk, which can be included in programs written in BASIC or other high level languages. This version is the minimum system for interactive graphics picture creation. 24K memory is required. This program and the next, take all the hard work out of creating graphics on the H19 or H89--an excellent product to have if you're into graphics. Some features include complete cursor position, Graphics, Reverse video, Graphics recall, Auto graphic next sequence, disk read to screen and disk write from screen, and many more. \$24.95.

Along the same line is VIDEO ARTIST PLUS, for \$39.95. It's a beefed up version of the Video Artist to allow creation and easy modification of graphics drawing. Fast operation, extensive command list, and the ability to create a picture segment disk library make this the best choice for programmers with a need to add graphics displays to their applications programs. 32K of memory is required for this one. Additional features include Scroll Screen any direction, with full wraparound, Cut and Paste, Insert character and line, delete character and line, 8 cursor motion directions, and more.

SCRIBE is a screen oriented text editor with many useful and unique commands. The text being edited is visible on the screen and changes to the text are instantly displayed. What you see is what you get! The 25th line displays a command menu so you do not have to remember all the commands. There are over 40 powerful commands for text manipulating and I/O. It will handle lines up to 255 characters, and has the ability to append text. Complete cursor control; character, line, screen, and memory. Left and Right justification and margin setting, actually make this a text processor, not just an editor. \$39.95. Requires 32-48K memory.

All of the above can be ordered as sample DEMO programs for \$6 each which includes documentation. This will allow you to sample it and see if you like it before shelling out the full amount.

Ron of NEWLINE tells me more is underway. I took a few seconds to play with Video Artist Plus, and was impressed.

Bill Perry, President of TRIONYX ELECTRONICS, POB 5131, Santa Ana, CA 92704 writes: "Several of our customers have now obtained the color graphics board and are using it with our memory board without any problems.

"There seems to be widespread concern, however, about the capability of the H8 18V power supply to supply both the memory board and the graphics board.

"We are now offering a voltage converter kit for the M-H8 memory board which will convert the M-H8 to a +8 volt only memory. This voltage converter kit will consist of a tiny pc board and some parts and will sell for \$19.95. The converter will fit neatly on the M-H8 pc board in place of the existing +12 and -5 volt regulator circuits.

"The converter uses the +8 volt supply to generate both the +12 and -5 volts for the memory chips. The standard 3 voltage 4116 memory chip is fundamentally more reliable than the +5 volt only memory chip which will be used by the Heath Co. on their new 64K dynamic memory board. It is also less expensive and easier to obtain than the single voltage chip."

So there you have it, another potential problem solved by the engineers at TRIONYX, who pull through again!

LUCIDATA P-8080 PASCAL is now available for the HDOS system from POLYBYTES, 325 19th St. S. E., Cedar Rapids, IA 52403 for \$95 (\$25 for doc only).

Some features are full compatibility with all HDOS files and devices, Random and Sequential disk file access, Assembly language user function support, 16 bit integer and 9 digit floating point arithmetic. It comes with a comprehensive 60 page manual which describes those features of the P-8080 PASCAL which do not conform to the ISO standard. For more info, write POLYBYTES.

From dilithium Press, Box 606, Beaverton, OR 97075 (503) 243-1158 is another H8 book, H-8 Programming for Beginners, for \$8.95. It is a short programming course that will lead the reader step by step into the basics of computer programming. It covers mainly machine language. I plan to do a review on this book in a future H8SCOOP.

Don Long's CLOCK BOARD is finally here. This board contains the clock with 12 parallel I/O ports.

The clock will display time and date containing time (23:59:59), Month (01-12), Date (00-31), Year (00-99), and Day of Week (0-6). It contains self charging nicad battery for constant clock power. Features include two real time periodic interrupts, 12 bidirectional 8 bit I/O ports and comes complete with extensive documentation.

This board uses the newest technology 8255 PPI chip for maximum flexibility without loading the H8 buss. Each 8255 PPI chip has 24 I/O lines which are generally divided into 4 groups, the mode and direction of data transfer being totally software selectable. Three modes may be selected under program control: Mode 0-Basic I/O (continuous transfer), Mode 1- Strobed I/O with interrupt support, Mode 2- Bidirectional buss with interrupt support. Almost any interface application can be implemented without external logic with this card.

Because of the flexibility of the I/O there is no need to load the H8 buss any longer. Everything can be done via the Parallel I/O this card offers. Programs or actions can be executed or performed at any pre-programmed date and time.

Uses and applications include:

- * Print time and date on all listings
- * Time and log computer useage
- * Energy management
- * Real time control systems
- * Security systems
- * Data acquisition

The board requires 8V at 500mA and the buss loading is 1 LS TTL load. 20 I/O port locations are required for the clock and all ports which are user definable. The time base accuracy is +/- 20 ppm.

I have one of these beauties sitting in my H8, and it is an EXCELLENT product. Don had some delays, but it was worth the wait. The layout is very good, and the features are unexcelled by any computer board presently in existence! Price? Assembled and tested board with clock and 12 parallel ports is only \$235. Don only has 29 of these left, so if you are interested, don't wait! If the demand is there, more may be made up. If you want the board with clock only, \$175; Board with clock and 3 parallel I/O ports, \$190; Board with 12 parallel I/O ports only, \$200. All these are assembled and tested units, and prices are INTRODUCTORY prices.

For ordering, or more information, contact Donald Long, of Environmental Control Systems, 9319 Willowview Lane, Houston, TX 77080. (713) 464-9315.

The DIABLO 630 is available for \$1995 from Omega Sales Co, 12 Meeting ST, Cumberland, RI 02864. They have a toll free number 1-800-556-7586, and accept VISA and MC. Before you get too excited, however, consider service. Let the buyer beware.

Charles Bennett, 6908 W 98 St, Overland Park, KS 66212, has available under MBASIC an INVENTORY/CUSTOMER/INVOICE Program set which sells for \$49.95. Random file I/O techniques are used. The inventory program will hold 900 items on a data disk, and the customer program will hold 750 customer files.

From James Czebiniak, 199 Vly Rd. Ext, Schenectady, NY 12309 (518) 869-8723 comes BASGEN, a program written in MBASIC that allows you to have form fill out capability on your H19/H89 using protected screens and random file I/O. Do field statements confuse you? BASGEN will solve all of these problems for you. Just give BASGEN some parameters to describe your screen and data files, and BASGEN will write a complete interactive MBASIC program using random file I/O and protected field screens.

The documentation that comes with it is quite complete and even helps you learn protected screens and random access file I/O under MBASIC, which includes fields, although you do not need it since the program does it all for you automatically. If you want to see what it can do send \$7 for the very complete manual and documentation, including a sample generated program. The entire package for BASGEN is \$50, which includes an interactive parameter entry program that takes all the guesswork out of entering your parameters. Your H19 can be an "intelligent" terminal.

From JWS Engineering, Box 67, Lebanon, NJ 08833 (from Microcomputing Add February 1981 Pg 107) comes Heath H8-Z80 bare board (whatever that is supposed to be) for \$34.95.

Hank Labarbara called to tell me BITZERO, 1413 Redeker Rd, DesPlains, IL, 60016, offers for \$39.95, a mod to the disk drive to allow floppy diskettes to be turned for both sides to be used. Requires no holes on diskette itself. If you want to do this, BEWARE, there are pros and cons of

using both sides of a diskette, and I am presenting this item, not endorsing it in any way.

I found a lower priced source of 4116 memory chips, from February 1981 COMPUTER SHOPPER, page 25. 150 ns M58759-20, Mitsubishi equivalent of 4116 8 for \$28, and National equivalent MM5290N-3 200 ns chips, 8 for \$24. These are from Component Supply, Inc., 1771 Junction Ave, San Jose, CA 95112, (408) 295-7171.

Still from JDR Microdevices, Inc, 1101 South Winchester Blvd, San Jose, CA 95128 1-800-538-5000 are 100% guaranteed 200 ns 4116's, 8 for \$29.95.

Jeff Sohn called to tell me he visited the Milwaukee Wisconsin Heath store to find they had in stock for the H89 a card to plug in the buss to provide two parallel ports. This card comes complete with cables and disk software (what that's for I don't know), and sells for \$125.

BIBLIOGRAPHY UPDATE

Microcomputing, February 1981 tops the list again, with an article by Donald Skiff on page 95, Getting the Most From Your H8. Don talks about and clarifies various aspects of the H8 such as PAM-8, the Keypad, H8 Display, the H8-5 serial interface board, the H8-2 parallel board, and the H8-4 4 port serial board. The interface board discussion is mainly programming of the USART and ACE (which took it's place on the H8-4).

Also by D.C. Shoemaker on page 169 is a shortie on Heath Commands Revealed where he clarifies CAT/JGL and HDOS STAND-ALONE features.

The February BYTE on page 18 featured some feedback on the H89, which are two letters defending the H89 against a previous report which appeared in BYTE.

REQUESTS

William Goodwin, 312 Lamson, Grand Ledge, MI 48837 is working on cassette-based FIG-FORTH for his H8. William would like to hear from others that have a cassette FORTH or are interested in it.

Prof. Grant Gustafson, 2243 South Belaire Dr, Salt Lake City, UT 84109 writes in about ideas and wants for MODEM operation:

This one, related to MCS. Grant made a text file called LOGIN.CMD and a SUBMIT file:

```
MCS1=RUN MCS;SEND;LOGIN.CMD;CONVERSATION
```

The file LOGIN.CMD contains the lines normally typed to log on the DEC-20 computer.

Using DUMP, even a 1C can be sent via this file. He also keeps another file which re- attaches the job after the phone line kills it.

He says "John Merrill and I once talked about writing a subroutine to add onto MCS which would allow the originate end to operate the destination end as though it were an AT:, with the destination end keyboard disabled.

"The idea, however implemented, would allow my friend Richard to call up, look over the disks on my H89 and copy whatever he pleased. In reverse, he could give me his stuff, directly to my disks.

"H89COMM falls short of this. Any ideas on how to do it? Hope to read the answer in H8SCOOP."

I don't have any, but if any readers out there do, especially the many MODEM readers I have, please respond to Grant, or H8SCOOP.

Doug Arndt, 1111 5th Ave NE, Watertown, SD 57201 would like to hear from computer enthusiasts involved with interfacing computer to a machine shop environment, such as numerical control machines.

Doug is also trying to find a device driver for the H8-2 Parallel card which allows parity so the H10 paper tape printer will punch with parity. He is also interested in a device driver to drive an ASR-33 Teletype (20mA) so it punches with Parity. Can anyone help Doug out?

SSGT James Romoser, POB 168, AFELM NATO 5ATAF HQ2, APO NEW YORK 09453 is stationed in Northern Italy and wants to know if anyone else is located in this area (Verona area) with an H8 he could get together with.

WHO'S WHO

Don Long (see address in NEW STUFF) has perfected a device driver for running a selectric via closed loop (no software timing loop). Throughput is more than doubled. Don's Clock Board, PC-12 is required and the driver is processor time independent (2 or 4 Mhz). If anyone is interested in this, send Don a SASE. (This is actually a better way to run the selectric, as the clutch is not cycling continuously which causes wear and tear. If you have this card or are planning to get it and have a selectric, this is the way to go. --Henry)

From Steve McBride, 2355 Washington St, Eugene, OR 97405. "Here's something I hope might be of help to others. I have a COMPRINT 912 Printer. This device automatically wraps lines at 80 characters which causes a problem if you are printing text with exactly 80 characters per line. After the 80th character, the printer does a CR and then another when it receives the wrap CR from the device driver. You can reset the width in the device driver, but sooner or later you will find yourself with a blank line because of this problem." Steve figured a patch to fix this problem. If anyone has this printer or has a similar problem with another printer, contact Steve, and send him a SASE.

I was talking about the paper tiger 460 printer in back issues, supposed to be a letter quality dot matrix printer for RS-232 serial interface. This means you can use it for H8 or H89. Alfred Llorens purchased one and connected it to his H89 and sent me a sample print. It seems to be quite an impressive printer for that price, capable of printing various lines/inch printing. Next page is a sample of some print Alfred sent me.

12

FREE = 68 ; MEMORY = 19700

NEWSOLAR.BAS JUNK .BAS FREE .BAS
SOLAR .BAS SOLARDEC.BAS

or whatever would be your directory.

This FREE and MEMORY would be from MBASIC by the way.

```
1 F=0:J=0
2 J=PEEK(5120+J)-
3 IF J=0 THEN 5
4 F=F+2:GOTO 2
5 PRINT "FREE =";F;"; MEMORY =";FRE(1)
6 OPEN "I",3,"DIR.DIR"
7 LINE INPUT #3,A$:PRINT A$:IF NOT EOF(3)
  THEN 7
8 CLOSE #3
```

H9 GRAFIX MOD PROGRAM

A program submitted by Richard Melvin, 2103 First St., La Grande, OR 97850 which works with the H9 GRAFIX mod is printed, which demonstrates the ability of motion using the GRAFIX character set. It also allows for many variations in its running. It is a rather short and simple program and even has a surprising finish. "Herman" races after the cursor, trying to leap on it and gobble it up. But what fate awaits whom?

```
10 REM "HERMAN" THE WORM BY RICHARD A.
    MELVIN, LA GRANDE, OREGON
20 REM DONE IN EXT.B.H.B. ON THE H8/H9
    FOR 'GRAFIX' (PRODUCT OF NORTHWEST
30 REM COMPUTER PRODUCTS, VANCOUVER, WA)
40 DIM Z(30):REM GRAPHIC CHARACTER ARRAY
50 FOR I=0 TO 23:REM LOOP TO ESTABLISH
    GRAFIX VARIABLES
60 READ Z(I)
70 NEXT
80 REM DATA FIELD FOR THE GRAFIX CHAR.
90 DATA 191,192,192,247,208,192
100 DATA 245,240,192,244,248,192
110 DATA 240,250,192,224,250,193
120 DATA 192,250,195,192,234,203
130 P=232:REM PORT ADDRESS FOR H9 VIA
    H8-4 (250 FOR THE H8-5)
140 C1=8:REM CHR$(8) BACKSPACE
150 P1=2:REM TIME VARIABLE-CAN BE VARIED
    TO CHANGE SPEED OF THE ACTION
160 FOR X=1 TO 20:REM LENGTH OF RUN
170 FOR B=0 TO 23:REM LOOP TO OUT GRAPHICS
180 C=C+1:REM COUNTER FOR # FRAMES
    (CHARACTERS) TO DISPLAY AT A TIME
190 OUT P,Z(B):REM PUT GRAPHIC CHR ON CRT
200 PAUSE P1:REM DELAY FOR TIMING
210 IF C=3 THEN GOSUB 270:REM 3 FRAMES
    HAVE BEEN SHOT, TIME TO
220 REM:BACK UP AND SHOOT 3 MORE
230 NEXT B
240 PRINT SPC(4);:REM ERASE PREVIOUS
    FRAMES AND LEAP ONE MORE
250 NEXT X
260 PAUSE:GOTO 160:REM HALT IN ACTION, HIT
    RETURN TO START OVER
270 FOR L=1 TO 3:OUT P,C1:NEXT L:REM
    BACKSPACE LOOP
280 C=0:REM CLEAR FRAME COUNTER AND RET.
290 END
```

REPORT

RTTY COMMUNICATIONS PROCESSOR REVIEW

This review on the RTTY program written by Howard Nurse, was submitted to H8SCOOP by Skip Prinsen, 3611 Merrimac Ave, San Diego, CA 92117, and edited by Henry Fale, editor H8SCOOP.

The COMMSOFT RTTY COMMUNICATIONS PROCESSOR is an excellent work of programming which incorporates most of the features of all of the other software packages that are available for the Apple and the Radio Shack, TRS-80 computers.

It includes the features of the earlier package that is offered by HUG (885-1023) plus many more. The documentation with the disk is very good, having sections on how to "patch" changes into certain areas of the program if you did not like certain parts of the program.

Following are brief outlines of the chapters in the manual that comes with the disk. Ch 1 explains the use of the manual and the specs of the program. Ch 2 covers the required system configurations needed to be able to use the program (H8/H17/H19 with H8-4 and 24K, or H89/H88-3 with 32K RAM. Either system requires HDOS). Ch 3 is system start-up procedures. Ch 4 is Initialization, which is great in that you do not have to redo the Initialization process each time that you call up the program. Just input the time and go to operate. Ch 5 explains the operation of the system. A word of warning here, READ the manual and then READ it again! You can in no way learn all of the control codes in one reading and I had to read the manual many times to get all the features that I wanted to use down pat. Ch 6, Interfacing and hardware. This section is very well laid out so that even a beginner will have little trouble in getting his system up and running. This section covers transmitters and receivers. It explains how to interface your computer to your terminal unit and it even gives a brief survey of the more popular amateur T.U's and the pros and cons of each one. Ch 7, the appendices give instructions on how to change port assignments, WRU response time, changing character speed tables,

changing the buffer empty diddle count, and the location in memory of the ASCII/BAUDOT tables so you can change it for copying such things as weather bulletins.

The software has split screen operation with a selectable size for the received and transmitted data. It uses the 24 lines of the terminal, and the 25th line is used to keep the time and date and other operating data in front of you at all times. Some of the features of this 25th status line is 24 hour time, date, left margin status, righthand justify status, BAUD rate, system mode (transmit/receive/CW id, buffer contents counter, cursor position counter, line counter, logging device (SY0:/SY1:/LP:), log status, and also log sectors remaining, if using disks to record to.

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The split screen allows copying off the air and also to be able to type ahead data while receiving. The screen is 80 characters wide max, of which you can only use 72 for outputting data. Another feature that is not good is that it will not give good picture results on overline pix so for the pix buff this can be a pain.

All of your data received and/or transmitted can be put to any of the device drivers you wish to use. If the frequency that you monitor is very busy you can have just traffic that is meant for your station saved to disk, thus cutting down on the read time. While data is being output to the disk, LP, or whatever, no data is lost, as it is put into a buffer and will update the screen when the output data is finished.

There are three permanent messages for testing and calling purposes; "the quick brown fox", "ryryryry", and "CQ CQ CQ" strings. Also there are three user programable buffers 70 characters in length that can be programmed to whatever you wish to have them set-up for.

Transmitter/Receiver control can be automatically controlled by the computer which is required if the "WRU" feature is to be used.

RTTY89 includes: A 73 page manual covering all aspects of amateur RTTY, a prompt card to help when running the program a vinyl disk storage sheet, the RTTY89.ABS distribution disk, a three ring binder to hold the package, and free updates for one year after registration is returned. Cost of the complete package is \$100.

My overall impression of this package is nothing but great. I recommend it highly to all. One thing to remember is that it will take several days use to get the hang of the control features of the program so you can get it to do what you want.

AT PRESS TIME

Last minute news I thought I'd throw in which I just received at press time. The new release of MBASIC should be in your hands by now. After taking so long to get it out, I understand there's numerous bugs in this release, so BEWARE.

HDOS version 2.0 is now being mailed out, and if you are on the list, you should be getting your update soon. I understand it's DYNAMITE.

If you have the Siemens disk drives, even the older ones, and need parts or service, this should help you. Siemens will supply parts and service to these drives. For parts, the number is (714) 991-9700. 240 E. Palias, Anaheim CA 92805. I guess for \$10 you can get a maintenance manual for your drives. For service, (714) 634-1914; 1601 S. Sinclair, Anaheim CA 92806.

NEXT MONTH I will start a series on interfacing the computer to the outside world, via a parallel port.

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\$12/Year (\$18 overseas AIRMAIL)--first year back issues.

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FOR SALE--2-32K memory boards and CPU board. Build a 64K H-8 for a minimum cost. Call or write for prices. William Hall, 5621 Maple Heights, Pittsburgh, PA 15232. (412) 683-4742.

FOR SALE--H8-5 Serial/Cassette Interface board for \$70. Thomas Bessire, 5817 Boca Raton, Apt. J108, Ft. Worth, TX 76112. (817) 451-3912.

FOR SALE--Assembled, operational; Heath H8 with H8-4 four port serial board, full 64K Trionyx memory (150ns 4116's), PAMGO ROM, dual drive H17 with HDOS, MBASIC, and many HUG and other disks. System includes H19 video terminal, with covers for all. First \$2000 or best offer, I'll ship. Eric Sturm, Mountain View Mobile Park, Lot B-4, Denver, CO 80230. (303) 399-6867.

WANTED--Heath weather station; not necessarily the digital kind. Ray Albrektson, 3265 N "G", San Bernardino, CA 92405. (I guess after Ray's recent house fire, he wants to keep an eye on the wind direction from now on!--Henry)

JOB POSITION AVAILABLE--We have a permanent full-time position open for an experienced programmer familiar with the Heathkit H8 computer. We need software support in developing a wide variety of interesting hardware products for the H8 computer. Interested persons should contact Bill Perry of Trionyx Electronics at (714) 556-8225.

FOR SALE--1 12K godbout memory board for \$50. Donald Long, 9319 Willowview Ln, Houston, TX 77080. (713) 464-9315.

FOR SALE--WH8-16 Heath Factory assembled 16K MEMORY in perfect condition. Reason for selling--have single Trionyx board and need to release a slot to make room for expansion. \$175 postpaid. Lloyd Dinkins, Box 22642, Memphis, TN 38122.

Henry